

Name: _____ ID: _____ Grade (A/B/C): _____ by _____ Grade (A/B/C): _____ by _____

Worksheet: Reinforcement Learning & Q-Learning

Problem 1: Discounted Returns & Recursion

Problem Setup: An agent operates in an episodic environment with a discount factor of $\gamma = 0.9$. Starting from time $t = 0$, it receives the following sequence of immediate rewards before the episode terminates at time step $T = 6$:

$$R_1 = +10, \quad R_2 = -5, \quad R_3 = +20, \quad R_4 = +5, \quad R_5 = -10, \quad R_6 = +15$$

Given Equations:

- Total Discounted Return: $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$
- Recursive Relationship: $G_t = R_{t+1} + \gamma G_{t+1}$

Questions: Calculate the exact total discounted return G_t for the following time steps. Show your work in the space provided and write your final answer on the line.

(a) $t = 5$

Calculation: Final Answer: $G_5 =$ _____

(b) $t = 3$

Calculation: Final Answer: $G_3 =$ _____

(c) $t = 2$

Calculation: Final Answer: $G_2 =$ _____

(d) $t = 0$

Calculation: Final Answer: $G_0 =$ _____

Problem 2: Q-Learning (Off-Policy TD Update)

Problem Setup: An agent is training using tabular Q-learning (using a Q-table) with a discount factor $\gamma = 0.8$ and a learning rate $\alpha = 0.25$. The agent transitions from State A to State B by taking action "East", receiving a reward of +4.0. Transition: ($S = A, \quad A = \text{East}, \quad R = +4.0, \quad S' = B$). The current Q-table has the following estimates:

$$Q(A, \text{East}) = 6.0, \quad Q(B, \text{North}) = 5.0, \quad Q(B, \text{South}) = 2.0, \quad Q(B, \text{West}) = 7.5$$

Given Equations (Q-Learning):

- TD Target = $R + \gamma \max_{a'} Q(S', a')$
- TD Error (δ) = TD Target - $Q(S, A)$
- Update Rule: $Q(S, A) \leftarrow Q(S, A) + \alpha \cdot \delta$

Questions: Calculate the Temporal-Difference (TD) Target for this transition. Next, calculate the TD Error (δ). Finally, calculate the new updated value for $Q(A, \text{East})$. Show your work in the space provided and write your final answer on the line.

(a) TD Target

Calculation: Final Answer: TD Target = _____

(b) TD Error (δ)

Calculation: Final Answer: δ = _____

(c) Q-Value Update

Calculation: Final Answer: $Q(A, \text{East}) =$ _____